

B.COM. (ACCOUNTING &amp; FINANCE) · SEMESTER V

# Corporate Financial Reporting – I

## Practical Question Bank with Detailed Solutions

### Module 4: Ind AS 16 – Property, Plant and Equipment

12 practical questions · Single, dual and multi-concept

#### HOW THIS QUESTION BANK IS BUILT

| Level   | What it tests                                                      | Questions | Marks     |
|---------|--------------------------------------------------------------------|-----------|-----------|
| Level 1 | Single concept: one rule at a time                                 | Q1–Q4     | 5–8 each  |
| Level 2 | Two concepts combined in one problem                               | Q5–Q8     | 8–10 each |
| Level 3 | Multi-concept, university exam pattern with sub-parts (5+5+5, 8+7) | Q9–Q12    | 15 each   |

#### HOW TO USE IT

- Attempt each **problem first** on your own. Then read the solution.
- Every solution has **step-by-step working tables** and journal entries wherever the standard needs them.
- The blue “**Why?**” box explains the logic in plain words. The amber “**Watch out**” box lists common mistakes and exam tips. The green box gives the **final answer** for quick checking.
- Use the **concept coverage map** to revise a specific topic. Use the **quick-reference sheet** just before the exam.

#### GROUND RULES FOR ALL QUESTIONS

- Amounts are in ₹. The financial year runs from 1 April to 31 March.
- GST is **recoverable** (input tax credit available) unless a question says otherwise. Income tax and deferred tax are ignored.
- Depreciation for part of a year is charged **month-wise**, from the date the asset is available for use to the date of disposal.
- Each question follows Ind AS as notified in India. Where Ind AS differs from IFRS, a note says so.

## Index of Questions

| Q  | Level | Marks | Topic                         | Concepts tested                                       |
|----|-------|-------|-------------------------------|-------------------------------------------------------|
| 1  | L1    | 5     | Cost of PPE                   | Recognition, directly attributable costs, exclusions  |
| 2  | L1    | 8     | Depreciation methods          | SLM, WDV, units of production, part-year              |
| 3  | L1    | 6     | Revaluation increase          | Revaluation surplus, OCI, depreciation                |
| 4  | L1    | 7     | Derecognition                 | Sale, retirement, destruction, insurance              |
| 5  | L2    | 8     | Decommissioning obligation    | Present value, unwinding, depreciation                |
| 6  | L2    | 10    | Components and replacement    | Component depreciation, derecognising a part          |
| 7  | L2    | 10    | Revaluation up and down       | OCI vs profit or loss, reversal rules                 |
| 8  | L2    | 8     | Changes in estimates          | Prospective change, method change, error              |
| 9  | L3    | 15    | Self-constructed plant        | Borrowing costs, testing proceeds, depreciation start |
| 10 | L3    | 15    | Subsequent expenditure        | Replacement, overhaul, spares, capacity addition      |
| 11 | L3    | 15    | Exchange and deferred payment | Commercial substance, cash price equivalent           |
| 12 | L3    | 15    | Revaluation to disposal       | Surplus transfer, derecognition, disclosure           |

## Concept Coverage Map

| Ind AS 16 concept                                                | Questions        |
|------------------------------------------------------------------|------------------|
| Recognition criteria and spare parts                             | 1, 10            |
| Initial measurement: purchase price, directly attributable costs | 1, 9             |
| Borrowing costs and self-constructed assets                      | 9                |
| Decommissioning / dismantling obligations                        | 5                |
| Exchange of assets and deferred payment                          | 11               |
| Depreciation methods and part-year depreciation                  | 2, 4, 9, 10      |
| Changes in estimate, method and prior-period error               | 8                |
| Component accounting                                             | 6, 10            |
| Subsequent expenditure and replacement                           | 6, 10            |
| Cost model vs revaluation model                                  | 2, 3, 7, 12      |
| Revaluation increase and decrease, OCI and profit or loss        | 3, 7             |
| Transfer of revaluation surplus to retained earnings             | 12               |
| Derecognition, gain or loss, insurance compensation              | 4, 6, 10, 11, 12 |
| Disclosures                                                      | 12               |

## Quick Reference – Ind AS 16 PPE

**Core rule:** recognise PPE when (i) future economic benefits are **probable** and (ii) cost can be **measured reliably**. Initial measurement is at cost.

| Cost =                                                                                    | Includes                                                                                                                                                                                                                                | Does not include                                                                                              |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Purchase price and directly attributable costs, plus initial estimate of dismantling cost | Duties and non-refundable taxes, freight, installation, site preparation, professional fees, testing costs (net of sale proceeds of test items in India), borrowing costs on qualifying assets, present value of dismantling obligation | Opening costs, advertising, training, general overheads, start-up losses, abnormal wastage, recoverable taxes |

| Topic                | Rule to remember                                                                                                                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Models               | Cost model: cost – accumulated depreciation – impairment. Revaluation model: fair value – later depreciation and impairment; revalue the whole class regularly                                                             |
| Revaluation increase | OCI (revaluation surplus), unless it reverses an earlier loss on the same asset charged to profit or loss                                                                                                                  |
| Revaluation decrease | Profit or loss, unless covered by a surplus on the same asset (then OCI)                                                                                                                                                   |
| Surplus transfer     | May be transferred to retained earnings as the asset is used or on derecognition. Never through profit or loss                                                                                                             |
| Depreciation         | (Cost – residual value) allocated over useful life. Starts when available for use. Stops at the earlier of held-for-sale classification and derecognition. Land is not depreciated                                         |
| Methods              | SLM, diminishing balance, units of production. Choose the one reflecting expected consumption of benefits                                                                                                                  |
| Review               | Useful life and residual value reviewed at least each year-end. Changes in estimate or method are <b>prospective</b> (Ind AS 8)                                                                                            |
| Components           | Depreciate significant parts separately. Replace a part: capitalise new, derecognise old                                                                                                                                   |
| Subsequent costs     | Capitalise if recognition criteria are met. Day-to-day servicing is expensed                                                                                                                                               |
| Derecognition        | On disposal or when no future benefits are expected. Gain or loss = net proceeds – carrying amount. Gains are not revenue                                                                                                  |
| Compensation         | Compensation from third parties for impaired, lost or given-up items is recognised in profit or loss when receivable                                                                                                       |
| Disclosures          | Measurement bases, methods, useful lives, gross amount and accumulated depreciation, reconciliation of carrying amount, pledges, commitments; for revalued assets the date, valuer, methods and cost-model carrying amount |

## Module 4 · Ind AS 16 – Property, Plant and Equipment

This module practises recognition and initial measurement of PPE, depreciation methods, the cost and revaluation models, component accounting, subsequent expenditure, decommissioning obligations, exchanges, changes in estimates, derecognition and disclosures.

**Level 1** (Q1–Q4) drills one concept at a time. **Level 2** (Q5–Q8) joins two concepts. **Level 3** (Q9–Q12) mixes several concepts in 15-mark exam-style questions.

### LEVEL 1 – SINGLE CONCEPT PRACTICE (Q1 to Q4)

#### Q1. Recognition and Cost of an Item of PPE

**Level 1 · Single concept** | **5 marks** | Concepts tested: recognition criteria, purchase price, directly attributable costs, costs that are not capitalised

#### PROBLEM

Rahul Industries Ltd. bought a new machine on 1 April 2025. Related information:

| No. | Particulars                                                    | ₹         |
|-----|----------------------------------------------------------------|-----------|
| 1   | List price of the machine                                      | 40,00,000 |
| 2   | Trade discount of 5% on list price                             | —         |
| 3   | Import duty (non-refundable)                                   | 3,00,000  |
| 4   | GST @ 18% on the net price (fully recoverable as input credit) | —         |
| 5   | Freight and insurance to bring the machine to the factory      | 1,20,000  |
| 6   | Site preparation (foundation and electrical connections)       | 2,50,000  |
| 7   | Installation and assembly                                      | 1,00,000  |
| 8   | Fees of engineers for supervising installation                 | 80,000    |
| 9   | Cost of testing whether the machine functions properly         | 50,000    |
| 10  | Net proceeds from sale of samples produced during testing      | 15,000    |
| 11  | Training of staff to operate the machine                       | 40,000    |
| 12  | General overheads allocated                                    | 60,000    |
| 13  | Initial operating losses while demand builds up                | 90,000    |
| 14  | Advertising to launch products made on the machine             | 1,00,000  |

**Required:** Compute the cost at which the machine is recognised and explain each item that is left out.

#### SOLUTION

##### Step 1 – Check recognition: does the item meet both criteria?

(i) It is **probable that future economic benefits** will flow to the company, and (ii) the **cost can be measured reliably**. The machine meets both, so it is recognised as PPE.

**Step 2 – Compute cost = purchase price + directly attributable costs**

| Particulars                                  | ₹                |
|----------------------------------------------|------------------|
| List price                                   | 40,00,000        |
| Less: Trade discount 5%                      | (2,00,000)       |
| <b>Net purchase price</b>                    | <b>38,00,000</b> |
| Add: Import duty (non-refundable)            | 3,00,000         |
| Add: Freight and insurance                   | 1,20,000         |
| Add: Site preparation                        | 2,50,000         |
| Add: Installation and assembly               | 1,00,000         |
| Add: Engineers' fees                         | 80,000           |
| Add: Cost of testing                         | 50,000           |
| Less: Net proceeds from sale of test samples | (15,000)         |
| <b>Cost of machine</b>                       | <b>46,85,000</b> |

**Step 3 – Items not included in cost**

| Item                     | ₹        | Reason                                                                       |
|--------------------------|----------|------------------------------------------------------------------------------|
| GST @ 18%                | 6,84,000 | Recoverable, so it is an asset (input credit), not part of cost              |
| Staff training           | 40,000   | Expense. It does not bring the machine to the location and condition needed. |
| General overheads        | 60,000   | Expense. Not directly attributable to the machine.                           |
| Initial operating losses | 90,000   | Expense. These arise after the machine is ready for use.                     |
| Advertising              | 1,00,000 | Expense. It is a cost of promoting products, not of the machine.             |

**Final answer**

The machine is recognised at **₹46,85,000**. Items 4, 11, 12, 13 and 14 are excluded.

**Why? – Simple explanation**

PPE is measured at **everything needed to get the asset ready to work the way management intends**: the price paid, taxes you cannot recover, delivery, installation, site preparation and testing.

Costs that arise **after** the asset is ready (training, launch advertising, start-up losses) or that are not specific to the asset (general overheads) are running costs of the business and go to profit or loss.

Net proceeds from selling items produced while testing are deducted from the cost of testing. Under Ind AS 16 as notified in India (2022 amendment, effective from 1 April 2022), the net proceeds are deducted from directly attributable costs and are not credited to profit or loss. This is a **carve-out from the IFRS treatment (IAS 16)**, where such proceeds and the related cost go to profit or loss.

**Watch out – Common mistakes / Exam tip**

- Trade discounts are deducted; recoverable GST is never added.
- Spare parts and servicing equipment are inventory unless they are major spares expected to be used for **more than one period**, in which case they are PPE.
- Recognition asks for two things, not one: probable future benefits **and** reliable measurement.

## Q2. Depreciation Methods – SLM, WDV, Units of Production

**Level 1 · Single concept | 8 marks** | Concepts tested: depreciable amount, straight-line, diminishing balance, units of production, part-year depreciation

### PROBLEM

Sun Textiles Ltd. bought a machine for ₹16,00,000 on 1 April 2025 and it was ready for use on that date. Estimated residual value is ₹1,00,000 and useful life is 4 years. The machine is expected to produce 3,00,000 units in total: 1,00,000; 80,000; 70,000 and 50,000 units in the four years.

**Required:** Calculate depreciation for each year under (a) straight-line method, (b) written-down-value method at 50% p.a. and (c) units-of-production method. (d) How would straight-line depreciation change if the machine was ready for use on 1 October 2025?

### SOLUTION

**Depreciable amount** = Cost – Residual value = 16,00,000 – 1,00,000 = ₹15,00,000.

#### (a) Straight-line method (SLM)

Annual depreciation = 15,00,000 ÷ 4 years = ₹3,75,000 every year.

#### (b) Written-down-value method (WDV) at 50%

Depreciation each year = 50% of the opening carrying amount. (The rate is chosen so the machine reaches its residual value at the end of year 4.)

#### (c) Units-of-production method

Depreciation per unit = 15,00,000 ÷ 3,00,000 units = ₹5 per unit.

#### Summary of all three methods

| Year         | SLM (₹)          | WDV (₹)          | WDV closing carrying amount (₹) | Units           | UoP (₹)          |
|--------------|------------------|------------------|---------------------------------|-----------------|------------------|
| FY 2025-26   | 3,75,000         | 8,00,000         | 8,00,000                        | 1,00,000        | 5,00,000         |
| FY 2026-27   | 3,75,000         | 4,00,000         | 4,00,000                        | 80,000          | 4,00,000         |
| FY 2027-28   | 3,75,000         | 2,00,000         | 2,00,000                        | 70,000          | 3,50,000         |
| FY 2028-29   | 3,75,000         | 1,00,000         | 1,00,000                        | 50,000          | 2,50,000         |
| <b>Total</b> | <b>15,00,000</b> | <b>15,00,000</b> |                                 | <b>3,00,000</b> | <b>15,00,000</b> |

All three methods depreciate exactly ₹15,00,000 in total. Only the **timing** differs.

#### (d) Machine ready for use on 1 October 2025 (SLM)

| Year                     | Period                              | Depreciation (₹) |
|--------------------------|-------------------------------------|------------------|
| FY 2025-26               | 1 Oct 2025 – 31 Mar 2026 (6 months) | 1,87,500         |
| FY 2026-27 to FY 2028-29 | Full years (3 years × ₹3,75,000)    | 11,25,000        |
| FY 2029-30               | 1 Apr 2029 – 30 Sep 2029 (6 months) | 1,87,500         |
| <b>Total</b>             |                                     | <b>15,00,000</b> |

#### Final answer

SLM ₹3,75,000 p.a.; WDV ₹8,00,000 / 4,00,000 / 2,00,000 / 1,00,000; UoP ₹5,00,000 / 4,00,000 / 3,50,000 / 2,50,000. Part-year SLM first year = ₹1,87,500.

**Why? – Simple explanation**

Depreciation spreads the **depreciable amount** over the asset's useful life. The method should mirror how the business expects to **use up** the asset's benefits.

**SLM** assumes equal use every year. **WDV** assumes the asset gives more benefit early on. **UoP** ties depreciation to actual output, which suits machines whose use varies.

Depreciation begins when the asset is **available for use** (ready to operate as management intends), not when it is first used.

**Watch out – Common mistakes / Exam tip**

- Residual value is deducted for SLM and UoP. For WDV you apply the rate to the carrying amount and the residual value is reached through the choice of rate.
- **Land is not depreciated** as it has an unlimited life.
- The method, useful life and residual value are reviewed **at least at each financial year-end**.
- In India, Schedule II to the Companies Act, 2013 gives indicative useful lives. A company may use a different life if it can justify and disclose it.



### Q3. Revaluation Model – Revaluation Increase

**Level 1 · Single concept | 6 marks** | Concepts tested: revaluation model, revaluation surplus in OCI, restating accumulated depreciation, depreciation after revaluation

#### PROBLEM

Orbit Ltd. bought plant on 1 April 2023 for ₹20,00,000. Useful life is 10 years, residual value nil, straight-line depreciation. The company follows the **revaluation model**. At **31 March 2026** the fair value of the plant is ₹16,80,000.

**Required:** (a) Pass the journal entry for the revaluation using (i) the proportionate (gross carrying amount) method and (ii) the elimination (net) method. (b) Compute depreciation for FY 2026-27.

#### SOLUTION

##### Step 1 – Carrying amount before revaluation

| Particulars                                    | ₹                |
|------------------------------------------------|------------------|
| Cost                                           | 20,00,000        |
| Accumulated depreciation (3 years × ₹2,00,000) | (6,00,000)       |
| <b>Carrying amount at 31 March 2026</b>        | <b>14,00,000</b> |
| Fair value                                     | 16,80,000        |
| <b>Revaluation increase</b>                    | <b>2,80,000</b>  |

##### Step 2 – Journal entry: (i) proportionate method

Fair value is  $16,80,000 \div 14,00,000 = 1.2$  times the carrying amount, so both cost and accumulated depreciation are increased by 20%.

|                              | Before (₹)       | After (₹)        | Change (₹)      |
|------------------------------|------------------|------------------|-----------------|
| Gross carrying amount (cost) | 20,00,000        | 24,00,000        | 4,00,000        |
| Accumulated depreciation     | 6,00,000         | 7,20,000         | 1,20,000        |
| <b>Carrying amount</b>       | <b>14,00,000</b> | <b>16,80,000</b> | <b>2,80,000</b> |

| Particulars                       | Dr (₹)   | Cr (₹)   |
|-----------------------------------|----------|----------|
| Plant – gross carrying amount A/c | 4,00,000 |          |
| To Accumulated depreciation A/c   |          | 1,20,000 |
| To Revaluation surplus (OCI) A/c  |          | 2,80,000 |

##### Step 2 – Journal entry: (ii) elimination method

| Particulars                                                                                                | Dr (₹)   | Cr (₹)   |
|------------------------------------------------------------------------------------------------------------|----------|----------|
| Accumulated depreciation A/c                                                                               | 6,00,000 |          |
| To Plant A/c                                                                                               |          | 6,00,000 |
| <i>(Being accumulated depreciation eliminated against the gross carrying amount; plant now ₹14,00,000)</i> |          |          |
| Plant A/c                                                                                                  | 2,80,000 |          |
| To Revaluation surplus (OCI) A/c                                                                           |          | 2,80,000 |
| <i>(Being plant restated to fair value of ₹16,80,000)</i>                                                  |          |          |

##### Step 3 – Depreciation for FY 2026-27 (part b)

Remaining life =  $10 - 3 = 7$  years. Depreciation =  $16,80,000 \div 7 = ₹2,40,000$ , compared with ₹2,00,000 before revaluation. The extra ₹40,000 a year is charged to profit or loss.

**Final answer**

Revaluation surplus of ₹2,80,000 is recognised in **other comprehensive income** and accumulated in equity. Depreciation for FY 2026-27 is ₹2,40,000.

**Why? – Simple explanation**

Under the **revaluation model** the asset is shown at fair value at the revaluation date, less later depreciation. The gain is not a realised profit, so it does not go to profit or loss; it is recorded in **OCI** and kept in equity as a “revaluation surplus”.

Either way of adjusting accumulated depreciation gives the same carrying amount (₹16,80,000) and the same surplus (₹2,80,000). Only the presentation of gross cost differs.

Depreciation after revaluation is based on the **new carrying amount** over the remaining life.

**Watch out – Common mistakes / Exam tip**

- If you revalue one item, you must revalue the **whole class** of PPE it belongs to (for example all plant and machinery). Cherry-picking is not allowed.
- Revaluations must be repeated regularly so the carrying amount does not differ materially from fair value.
- A revaluation increase is credited to profit or loss only to the extent it reverses an earlier loss on the same asset (see Q7).

## Q4. Derecognition – Sale, Retirement and Destruction

**Level 1 • Single concept | 7 marks** | Concepts tested: gain or loss on derecognition, depreciation up to disposal date, retirement, insurance compensation

### PROBLEM

Three independent situations of Global Traders Ltd.:

- **(a)** A machine bought on 1 April 2021 for ₹12,00,000 (useful life 10 years, nil residual value, straight-line) was **sold on 30 September 2025 for ₹8,50,000**. Dismantling and brokerage cost ₹20,000.
- **(b)** An old machine with cost ₹3,00,000 and accumulated depreciation ₹2,50,000 was **permanently retired** from use on 31 March 2026. It has no scrap value and no further benefits are expected.
- **(c)** A plant with carrying amount ₹5,00,000 was **destroyed by fire** on 10 February 2026. The insurer approved a claim of ₹4,00,000 on 20 March 2026.

**Required:** Compute the gain or loss and pass journal entries in each case.

### SOLUTION

#### (a) Sale of machine

##### Step 1 – Charge depreciation up to the date of sale

| Particulars                                                   | ₹               |
|---------------------------------------------------------------|-----------------|
| Annual depreciation (12,00,000 ÷ 10)                          | 1,20,000        |
| Accumulated depreciation up to 31 March 2025 (4 years)        | 4,80,000        |
| Depreciation 1 April 2025 – 30 September 2025 (6 months)      | 60,000          |
| Total accumulated depreciation at date of sale                | 5,40,000        |
| <b>Carrying amount at date of sale (12,00,000 – 5,40,000)</b> | <b>6,60,000</b> |

##### Step 2 – Gain or loss = Net disposal proceeds – Carrying amount

| Particulars                                           | ₹               |
|-------------------------------------------------------|-----------------|
| Sale proceeds                                         | 8,50,000        |
| Less: dismantling and brokerage                       | (20,000)        |
| Net disposal proceeds                                 | 8,30,000        |
| Less: carrying amount                                 | (6,60,000)      |
| <b>Gain on disposal (profit or loss, not revenue)</b> | <b>1,70,000</b> |

| Particulars                     | Dr (₹)   | Cr (₹)    |
|---------------------------------|----------|-----------|
| Depreciation A/c                | 60,000   |           |
| To Accumulated depreciation A/c |          | 60,000    |
| Bank A/c (net proceeds)         | 8,30,000 |           |
| Accumulated depreciation A/c    | 5,40,000 |           |
| To Machine A/c                  |          | 12,00,000 |
| To Gain on disposal of PPE A/c  |          | 1,70,000  |

#### (b) Retirement of machine

Carrying amount = 3,00,000 – 2,50,000 = ₹50,000. There are no proceeds and no future benefits, so the whole carrying amount is a **loss of ₹50,000**.

| Particulars                                   | Dr (₹)   | Cr (₹)   |
|-----------------------------------------------|----------|----------|
| Accumulated depreciation A/c                  | 2,50,000 |          |
| Loss on retirement of PPE (Profit & Loss) A/c | 50,000   |          |
| To Machine A/c                                |          | 3,00,000 |

### (c) Destruction by fire

Two separate items: (1) the asset is derecognised and its carrying amount of ₹5,00,000 is a loss; (2) the insurance compensation is recognised as income when it **becomes receivable**. The insurer approved the claim in the same year, so both fall in FY 2025-26; the **net loss is ₹1,00,000**.

| Particulars                                   | Dr (₹)   | Cr (₹)   |
|-----------------------------------------------|----------|----------|
| Loss by fire (Profit & Loss) A/c              | 5,00,000 |          |
| To Plant A/c (carrying amount)                |          | 5,00,000 |
| Insurance claim receivable A/c                | 4,00,000 |          |
| To Insurance compensation (Profit & Loss) A/c |          | 4,00,000 |

### Final answer

(a) Gain ₹1,70,000. (b) Loss ₹50,000. (c) Loss ₹5,00,000 and compensation income ₹4,00,000 (net loss ₹1,00,000).

### Why? – Simple explanation

An asset is removed from the books when it is **sold** or when **no future benefits** are expected from it. The gain or loss is simply the difference between what you get and the carrying amount.

Depreciation must be charged up to the **date of disposal**; otherwise the carrying amount is too high and the gain is understated.

Insurance compensation is a **separate** event from the loss. It is recognised when the right to receive it arises, and the two are not netted in one line.

### Watch out – Common mistakes / Exam tip

- Gains on derecognition are **not revenue**. Show them as other income.
- If the claim were approved only after 31 March 2026, the ₹4,00,000 income would fall in the next year, and this year would show the full ₹5,00,000 loss.
- An asset that is temporarily idle is **not** derecognised; it continues to be depreciated. Assets held for sale (Ind AS 105) stop being depreciated.
- A company that routinely sells items it previously rented out transfers them to inventory at carrying amount when they are held for sale.

**LEVEL 2 – TWO CONCEPTS COMBINED (Q5 to Q8)****Q5. Decommissioning Obligation and Depreciation**

**Level 2 - Two concepts combined** | **8 marks** | Concepts tested: initial cost including dismantling obligation, present value, unwinding of discount, depreciation

**PROBLEM**

On **1 April 2025** Tara Power Ltd. installed a plant at a cost of **₹50,00,000**. The law requires the company to dismantle the plant and restore the site at the end of its useful life of **10 years**. The estimated cost at that time is **₹20,00,000**. The applicable discount rate is 8% p.a.; the present value factor for 10 years at 8% is **0.4632**. Residual value is nil; depreciation is straight-line.

**Required:** (a) Cost of the plant. (b) Journal entries on 1 April 2025 and at 31 March 2026. (c) Carrying amounts of the plant and the provision at 31 March 2026 and the provision at 31 March 2027.

**SOLUTION****Step 1 – Present value of the future dismantling cost**

PV = ₹20,00,000 × 0.4632 = **₹9,26,400**. The company owes ₹20,00,000 in ten years, but today's liability is only the discounted amount.

**Step 2 – Cost of the plant**

| Particulars                                            | ₹                |
|--------------------------------------------------------|------------------|
| Purchase and installation cost                         | 50,00,000        |
| Add: present value of dismantling and restoration cost | 9,26,400         |
| <b>Cost of plant</b>                                   | <b>59,26,400</b> |

**Step 3 – Journal entries**

| Particulars                                           | Dr (₹)    | Cr (₹)    |
|-------------------------------------------------------|-----------|-----------|
| <b>1 April 2025</b>                                   |           |           |
| Plant A/c                                             | 59,26,400 |           |
| To Bank A/c                                           |           | 50,00,000 |
| To Provision for dismantling A/c                      |           | 9,26,400  |
| <b>31 March 2026</b>                                  |           |           |
| Depreciation A/c (59,26,400 ÷ 10)                     | 5,92,640  |           |
| To Accumulated depreciation A/c                       |           | 5,92,640  |
| Finance cost – unwinding of discount (₹9,26,400 × 8%) | 74,112    |           |
| To Provision for dismantling A/c                      |           | 74,112    |

**Step 4 – Carrying amounts**

| Particulars                                          | ₹                |
|------------------------------------------------------|------------------|
| Plant: cost                                          | 59,26,400        |
| Less: depreciation for year 1                        | (5,92,640)       |
| <b>Carrying amount of plant at 31 March 2026</b>     | <b>53,33,760</b> |
| Provision at 1 April 2025                            | 9,26,400         |
| Add: unwinding of discount, year 1 (8%)              | 74,112           |
| <b>Provision at 31 March 2026</b>                    | <b>10,00,512</b> |
| Add: unwinding of discount, year 2 (₹10,00,512 × 8%) | 80,041           |
| <b>Provision at 31 March 2027</b>                    | <b>10,80,553</b> |

Charge to profit or loss in year 1 = depreciation ₹5,92,640 + finance cost ₹74,112 = **₹6,66,752**. By the end of year 10 the provision will have grown to about ₹20,00,000.

#### Final answer

Cost of plant **₹59,26,400**. Annual depreciation **₹5,92,640**. Provision: ₹9,26,400 → ₹10,00,512 → ₹10,80,553.

#### Why? – Simple explanation

When you install the plant, you also take on a **legal obligation** to clean up later. That obligation is a cost of having the plant, so its present value is added to the asset's cost and depreciated over the plant's life.

Because the payment is ten years away, the liability is recorded at **present value**. Each year the discount "unwinds" and is charged as a **finance cost**. It is not added to the asset again.

So the plant's cost is spread through depreciation and the interest on the obligation is spread through finance cost. Together they build up to the ₹20,00,000 that will eventually be paid.

#### Watch out – Common mistakes / Exam tip

- Do not add the undiscounted ₹20,00,000 to the plant cost.
- The provision is measured under Ind AS 37; only the initial estimate is added to PPE.
- Later changes in the estimate or discount rate adjust the asset cost and the provision (under the cost model), and the revised depreciable amount is depreciated over the remaining life.

## Q6. Component Accounting and Replacement of a Part

**Level 2 • Two concepts combined | 10 marks** | Concepts tested: components with different lives, depreciation by component, replacement, derecognition of replaced part, repairs

### PROBLEM

On **1 April 2025** Skyline Ltd. bought an office building. The **land** on which it stands cost ₹2,00,00,000. The building cost ₹6,00,00,000 with the following significant components, all with nil residual value:

| Component               | Cost (₹)           | Useful life (years) |
|-------------------------|--------------------|---------------------|
| Structure (RCC frame)   | 3,60,00,000        | 60                  |
| Lifts                   | 60,00,000          | 15                  |
| Air-conditioning (HVAC) | 90,00,000          | 10                  |
| Electrical and plumbing | 60,00,000          | 20                  |
| Roof waterproofing      | 30,00,000          | 5                   |
| <b>Total building</b>   | <b>6,00,00,000</b> |                     |

**Required:** (a) Annual depreciation using component accounting. (b) Depreciation if the entire building were treated as one asset with a 60-year life, and the difference. (c) On **31 March 2029** the roof waterproofing was replaced at ₹40,00,000 (new life 5 years); the same day ₹2,00,000 was spent on annual maintenance of the air-conditioning. Give the accounting treatment. (d) Depreciation on land?

### SOLUTION

#### (a) Depreciation by component (straight-line)

| Component               | Cost (₹)           | Life | Annual depreciation (₹) |
|-------------------------|--------------------|------|-------------------------|
| Structure (RCC frame)   | 3,60,00,000        | 60   | 6,00,000                |
| Lifts                   | 60,00,000          | 15   | 4,00,000                |
| Air-conditioning (HVAC) | 90,00,000          | 10   | 9,00,000                |
| Electrical and plumbing | 60,00,000          | 20   | 3,00,000                |
| Roof waterproofing      | 30,00,000          | 5    | 6,00,000                |
| <b>Total</b>            | <b>6,00,00,000</b> |      | <b>28,00,000</b>        |

#### (b) Single-asset approach

$600,00,000 \div 60 = ₹10,00,000$  per year. Component accounting gives ₹28,00,000, a difference of ₹18,00,000 per year, because short-life parts such as the roof and air-conditioning wear out much faster than the structure.

#### (c) Replacement of roof waterproofing on 31 March 2029

##### Step 1 – Derecognise the old roof

| Particulars                                    | ₹               |
|------------------------------------------------|-----------------|
| Cost of old roof                               | 30,00,000       |
| Accumulated depreciation (4 years × ₹6,00,000) | (24,00,000)     |
| <b>Carrying amount derecognised = loss</b>     | <b>6,00,000</b> |

**Step 2 – Capitalise the new roof and treat maintenance as an expense**

| Particulars                                           | Dr (₹)    | Cr (₹)    |
|-------------------------------------------------------|-----------|-----------|
| Accumulated depreciation – roof A/c                   | 24,00,000 |           |
| Loss on derecognition of old roof (Profit & Loss) A/c | 6,00,000  |           |
| To Building – roof (old) A/c                          |           | 30,00,000 |
| Building – roof (new) A/c                             | 40,00,000 |           |
| To Bank A/c                                           |           | 40,00,000 |
| Repairs and maintenance (Profit & Loss) A/c           | 2,00,000  |           |
| To Bank A/c                                           |           | 2,00,000  |

Depreciation on the new roof from FY 2029-30 =  $40,00,000 \div 5 = \text{₹}8,00,000$  per year.

**(d) Land**

Nil. Land has an unlimited useful life and is **not depreciated**. It is a separate asset even though the building stands on it.

**Final answer**

(a) **₹28,00,000** p.a. (b) ₹10,00,000 p.a., a difference of ₹18,00,000. (c) Loss on old roof **₹6,00,000**; new roof capitalised at **₹40,00,000**; maintenance ₹2,00,000 expensed. (d) Nil.

**Why? – Simple explanation**

If a building has parts that wear out at very different speeds, a single life would either over-depreciate the long-life structure or leave short-life parts on the books long after they are gone. So each **significant** part is depreciated over its own life.

When a part is replaced, the **old part comes out** of the books (with a loss if not fully depreciated) and the **new part goes in**, so the same part is never counted twice.

Routine servicing and maintenance only keep the asset working as before. They do not create a new asset, so they are **expenses**.

**Watch out – Common mistakes / Exam tip**

- If the original cost of the replaced part is not known, the cost of the replacement can be used as an indication of what the old part cost when it was bought.
- Components with the same life and depreciation method may be grouped together.
- Do not derecognise the old part **and** forget to remove its accumulated depreciation.



## Q7. Revaluation – Increase Followed by Decrease and Recovery

**Level 2 · Two concepts combined | 10 marks** | Concepts tested: revaluation model, revaluation surplus (OCI), revaluation loss (P&L), reversal rules, depreciation on revalued amount

### PROBLEM

Jaya Ltd. bought a plant on **1 April 2023** for **₹10,00,000** (useful life 10 years, nil residual value, straight-line). It follows the **revaluation model** and revalues at each year-end. Fair values (from an independent valuer):

| Date          | Fair value (₹) | Remaining life after revaluation |
|---------------|----------------|----------------------------------|
| 31 March 2024 | 10,80,000      | 9 years                          |
| 31 March 2025 | 7,20,000       | 8 years                          |
| 31 March 2026 | 7,35,000       | 7 years                          |

**Required:** Show, for each year, the depreciation, the effect of revaluation on OCI and profit or loss, the journal entries (elimination method) and the balance in revaluation surplus. Assume no annual transfer from revaluation surplus to retained earnings.

### SOLUTION

#### Step 1 – Rules to apply

- **Increase:** credit to OCI (revaluation surplus), **except** to the extent it reverses an earlier decrease on the same asset that was charged to profit or loss; that part goes to profit or loss.
- **Decrease:** charge to profit or loss, **except** to the extent of any revaluation surplus on the same asset; that part is debited to OCI.

#### Step 2 – Year-by-year working

|                                                             | FY 2023-24      | FY 2024-25      | FY 2025-26      |
|-------------------------------------------------------------|-----------------|-----------------|-----------------|
| Opening carrying amount                                     | 10,00,000       | 10,80,000       | 7,20,000        |
| Depreciation for the year                                   | 1,00,000        | 1,20,000        | 90,000          |
| <b>Carrying amount before revaluation</b>                   | <b>9,00,000</b> | <b>9,60,000</b> | <b>6,30,000</b> |
| Fair value                                                  | 10,80,000       | 7,20,000        | 7,35,000        |
| Revaluation increase / (decrease)                           | 1,80,000        | (2,40,000)      | 1,05,000        |
| – to OCI (revaluation surplus)                              | 1,80,000        | (1,80,000)      | 45,000          |
| – to profit or loss                                         | —               | (60,000)        | 60,000          |
| <b>Closing revaluation surplus</b>                          | <b>1,80,000</b> | <b>0</b>        | <b>45,000</b>   |
| Total effect on profit or loss (depreciation ± revaluation) | (1,00,000)      | (1,80,000)      | (30,000)        |

Depreciation: FY24 =  $10,00,000 \div 10 = 1,00,000$ ; FY25 =  $10,80,000 \div 9 = 1,20,000$ ; FY26 =  $7,20,000 \div 8 = 90,000$ .  
Negative figures in the last row are charges (expense).

#### Step 3 – Journal entries

| Particulars                      | Dr (₹)   | Cr (₹)   |
|----------------------------------|----------|----------|
| <b>31 March 2024</b>             |          |          |
| Depreciation A/c                 | 1,00,000 |          |
| To Accumulated depreciation A/c  |          | 1,00,000 |
| Accumulated depreciation A/c     | 1,00,000 |          |
| To Plant A/c                     |          | 1,00,000 |
| Plant A/c                        | 1,80,000 |          |
| To Revaluation surplus (OCI) A/c |          | 1,80,000 |
| <b>31 March 2025</b>             |          |          |

| Particulars                                                        | Dr (₹)   | Cr (₹)   |
|--------------------------------------------------------------------|----------|----------|
| Depreciation A/c                                                   | 1,20,000 |          |
| To Accumulated depreciation A/c                                    |          | 1,20,000 |
| Accumulated depreciation A/c                                       | 1,20,000 |          |
| To Plant A/c                                                       |          | 1,20,000 |
| Revaluation surplus (OCI) A/c                                      | 1,80,000 |          |
| Revaluation loss (Profit & Loss) A/c                               | 60,000   |          |
| To Plant A/c                                                       |          | 2,40,000 |
| <b>31 March 2026</b>                                               |          |          |
| Depreciation A/c                                                   | 90,000   |          |
| To Accumulated depreciation A/c                                    |          | 90,000   |
| Accumulated depreciation A/c                                       | 90,000   |          |
| To Plant A/c                                                       |          | 90,000   |
| Plant A/c                                                          | 1,05,000 |          |
| To Revaluation gain (Profit & Loss) A/c (reversal of earlier loss) |          | 60,000   |
| To Revaluation surplus (OCI) A/c                                   |          | 45,000   |

**Final answer**

Closing revaluation surplus at 31 March 2026 = ₹45,000. In FY 2024-25, ₹60,000 of the fall hits profit or loss; in FY 2025-26, that ₹60,000 is credited back and the balance ₹45,000 goes to OCI.

**Why? – Simple explanation**

Revaluation gains are unrealised, so they normally sit in OCI. Revaluation losses are recognised in profit or loss (prudence), except that a loss can first **use up** any surplus already created for the same asset.

If the asset later recovers, the loss that was charged to profit or loss is **credited back to profit or loss first** (it was an expense earlier, so the reversal is income now). Only the excess goes to OCI.

Each year, depreciation is charged on the latest fair value over the remaining life. That is why depreciation changes every year here.

**Watch out – Common mistakes / Exam tip**

- Surplus and losses are tracked **asset by asset** (or class by class as per policy); a surplus on one asset cannot absorb a loss on another.
- Do not credit the whole ₹1,05,000 to OCI in FY 2025-26; ₹60,000 must go first to profit or loss.
- If the company transfers excess depreciation to retained earnings each year (see Q12), the surplus balance would be lower.

## Q8. Change in Estimate, Change in Method and Error

**Level 2 • Two concepts combined | 8 marks** | Concepts tested: review of useful life, residual value and method, prospective change (Ind AS 8), difference from an error

### PROBLEM

Arya Ltd. bought a machine on **1 April 2022** for **₹15,00,000**. Useful life 10 years, residual value ₹1,50,000, straight-line. On **1 April 2025** the management reviews the machine and concludes:

- **(a)** The machine will now last only **4 more years** and the residual value will be **₹1,00,000**.
- **(b)** Alternatively, keeping other estimates unchanged, the company changes the method from straight-line to **written-down-value at 30%** because the machine now loses efficiency faster.
- **(c)** Separately, an auditor finds that depreciation of ₹1,35,000 for FY 2022-23 was **omitted by mistake**.

**Required:** Compute depreciation for FY 2025-26 under (a) and (b), and state how (c) is treated.

### SOLUTION

#### Step 1 – Carrying amount at 1 April 2025

| Particulars                                        | ₹                |
|----------------------------------------------------|------------------|
| Original depreciable amount (15,00,000 – 1,50,000) | 13,50,000        |
| Annual depreciation (13,50,000 ÷ 10)               | 1,35,000         |
| Accumulated depreciation, 3 years                  | 4,05,000         |
| <b>Carrying amount at 1 April 2025</b>             | <b>10,95,000</b> |

#### (a) Change in useful life and residual value

Apply the new estimates **prospectively** to the carrying amount: no restatement of earlier years.

| Particulars                                                     | ₹               |
|-----------------------------------------------------------------|-----------------|
| Carrying amount at 1 April 2025                                 | 10,95,000       |
| Less: revised residual value                                    | (1,00,000)      |
| Revised depreciable amount                                      | 9,95,000        |
| Remaining life                                                  | 4 years         |
| <b>Depreciation for FY 2025-26 and each of the next 3 years</b> | <b>2,48,750</b> |

Depreciation rises from ₹1,35,000 to ₹2,48,750, an increase of ₹1,13,750 per year.

#### (b) Change in method to WDV 30%

A change in depreciation method is a change in **accounting estimate** (it reflects a change in the expected pattern of consumption). Apply it prospectively: depreciation for FY 2025-26 =  $30\% \times ₹10,95,000 = ₹3,28,500$ . Earlier years are not changed.

#### (c) Omitted depreciation

This is a **prior-period error**, not a change in estimate. It is corrected **retrospectively**: restate the comparative figures (or the opening balances of the earliest period presented) as if the error had never happened, and disclose the nature of the error and the amount of correction.

| Feature                 | Change in estimate (a), (b)                  | Prior-period error (c)                                 |
|-------------------------|----------------------------------------------|--------------------------------------------------------|
| Cause                   | New information or a change in circumstances | Omission or misuse of information that was available   |
| Treatment               | Prospective: current and future periods      | Retrospective: restate comparatives / opening balances |
| Effect on earlier years | None                                         | Corrected                                              |

**Final answer**

(a) Depreciation ₹2,48,750 p.a. for 4 years. (b) ₹3,28,500 for FY 2025-26. (c) Correct retrospectively under Ind AS 8.

**Why? – Simple explanation**

Useful life, residual value and method are **estimates**. When the picture changes, you simply write off the remaining book value over the remaining life; the past was reasonable given what was known.

An error is different: the past was **wrong**. So the past is corrected.

**Watch out – Common mistakes / Exam tip**

- Useful life, residual value and the depreciation method must be reviewed **at least at each financial year-end**.
- A change in estimate is disclosed if it has a material effect in the current or future periods.
- Do not touch the ₹4,05,000 of depreciation already charged in (a) and (b).

**LEVEL 3 – MULTI-CONCEPT, EXAM PATTERN (Q9 to Q12)****Q9. Self-constructed Plant – Cost, Borrowing Costs and Depreciation**

**Level 3 · Multi-concept (exam pattern) | 15 marks |** Concepts tested: self-constructed asset, abnormal wastage, testing costs and sale proceeds, borrowing costs (Ind AS 23), date of depreciation start

**PROBLEM**

Zenith Ltd. constructed a plant for its own use. **Construction started on 1 April 2025**, the plant was **ready for its intended use on 31 December 2025**, and **commercial production began on 1 March 2026** (delayed by a pending statutory licence). Costs:

| No. | Particulars                                                                                      | ₹         |
|-----|--------------------------------------------------------------------------------------------------|-----------|
| 1   | Direct materials consumed (includes ₹3,00,000 abnormal wastage due to negligence)                | 60,00,000 |
| 2   | Direct labour                                                                                    | 24,00,000 |
| 3   | Site preparation                                                                                 | 6,00,000  |
| 4   | Supervision cost directly attributable to construction                                           | 5,00,000  |
| 5   | General administrative overheads allocated                                                       | 4,00,000  |
| 6   | Cost of testing whether the plant functions properly (including trial run)                       | 3,00,000  |
| 7   | Net proceeds from sale of sample items produced during testing                                   | 1,00,000  |
| 8   | Specific term loan of ₹1,00,00,000 at 9% p.a. taken on 1 April 2025, fully used for construction |           |
| 9   | Income earned from temporary investment of unspent loan money during construction                | 1,50,000  |

The plant has a useful life of **10 years** and a residual value of **₹25,000** (straight-line). The financial year ends on 31 March.

**Required:** (a) [8 marks] Compute the cost of the plant. (b) [7 marks] Compute depreciation for FY 2025-26 and list the items charged to profit or loss.

**SOLUTION****(a) Cost of the plant****Step 1 – Borrowing costs eligible for capitalisation (Ind AS 23)**

| Particulars                                                                                 | ₹               |
|---------------------------------------------------------------------------------------------|-----------------|
| Interest for construction period 1 Apr – 31 Dec 2025 $(1,00,00,000 \times 9\% \times 9/12)$ | 6,75,000        |
| Less: income from temporary investment of the specific loan                                 | (1,50,000)      |
| <b>Borrowing cost capitalised</b>                                                           | <b>5,25,000</b> |

Capitalisation **stops on 31 December 2025** when the plant is ready. Interest for 1 Jan – 31 Mar 2026  $(1,00,00,000 \times 9\% \times 3/12 = \text{₹}2,25,000)$  is a finance cost in profit or loss, even though the licence delay meant production had not started.

**Step 2 – Cost sheet**

| Particulars                                                              | ₹                |
|--------------------------------------------------------------------------|------------------|
| Direct materials $(60,00,000 - 3,00,000 \text{ abnormal wastage})$       | 57,00,000        |
| Direct labour                                                            | 24,00,000        |
| Site preparation                                                         | 6,00,000         |
| Directly attributable supervision                                        | 5,00,000         |
| Cost of testing, net of sale proceeds of samples $(3,00,000 - 1,00,000)$ | 2,00,000         |
| Borrowing costs capitalised                                              | 5,25,000         |
| <b>Cost of plant</b>                                                     | <b>99,25,000</b> |

**(b) Depreciation and items charged to profit or loss****Step 1 – Depreciation**

Depreciable amount = 99,25,000 – 25,000 = ₹99,00,000. Annual depreciation =  $\div 10$  = **₹9,90,000**. Depreciation starts when the plant is **available for use**, which is **1 January 2026** (not 1 March when production started). FY 2025-26 depreciation = ₹9,90,000  $\times 3/12$  = **₹2,47,500**.

**Step 2 – Items charged to profit or loss**

| Item                             | ₹                | Reason                                                           |
|----------------------------------|------------------|------------------------------------------------------------------|
| Abnormal wastage                 | 3,00,000         | Wasteful cost, not necessary to bring the asset to its condition |
| General administrative overheads | 4,00,000         | Not directly attributable to construction                        |
| Interest 1 Jan – 31 Mar 2026     | 2,25,000         | Asset is ready; capitalisation has ceased                        |
| Depreciation for 3 months        | 2,47,500         | Charged from date plant is available for use                     |
| <b>Total</b>                     | <b>11,72,500</b> |                                                                  |

**Final answer**

Cost of plant **₹99,25,000**. Depreciation FY 2025-26 **₹2,47,500**. Items to profit or loss: abnormal wastage ₹3,00,000, admin overheads ₹4,00,000, post-completion interest ₹2,25,000, depreciation ₹2,47,500.

**Why? – Simple explanation**

A self-constructed asset is recorded at the same cost principles as a bought one: **costs directly needed to build and prepare it**. Waste, general overheads and costs after completion are excluded.

Interest on money borrowed to build a “qualifying asset” is part of its cost while the asset is being made ready. The income earned by parking the borrowed money temporarily is deducted because it reduces the real interest cost.

The asset starts depreciating when it is **ready to operate**, whether or not the business has actually begun using it.

**Watch out – Common mistakes / Exam tip**

- **Ind AS vs IFRS:** Under Ind AS 16 as notified in India (effective 1 April 2022), net proceeds from sale of items produced during testing reduce the cost of the asset; under IAS 16 they would go to profit or loss along with the cost of producing them.
- Borrowing costs are capitalised only for the period when activity to prepare the asset is in progress. Once ready, stop.
- Marks: 8 for cost sheet with borrowing costs, 7 for depreciation start date and P&L items.

## Q10. Subsequent Expenditure – Capital or Revenue?

**Level 3 • Multi-concept (exam pattern) | 15 marks** | Concepts tested: recognition of subsequent costs, replacement of parts, major overhaul, spares, capacity enhancement, derecognition, depreciation

### PROBLEM

Sunrise Chemicals Ltd. bought a plant on **1 April 2021** for **₹1,20,00,000** (nil residual value, straight-line) with two components:

| Component      | Cost (₹)  | Useful life |
|----------------|-----------|-------------|
| Main structure | 96,00,000 | 12 years    |
| Furnace lining | 24,00,000 | 5 years     |

During **FY 2025-26** the following happened:

| No. | Event                                                                                                                                                                                                      | ₹         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1   | On 1 April 2025 the furnace lining was replaced with a new lining (new life 5 years)                                                                                                                       | 30,00,000 |
| 2   | Routine repairs and maintenance                                                                                                                                                                            | 4,00,000  |
| 3   | On 1 October 2025, mandatory major overhaul (required every 3 years for the plant to keep operating). The previous overhaul on 1 October 2022 cost ₹7,50,000, was capitalised and depreciated over 3 years | 9,00,000  |
| 4   | On 1 April 2025, a standby main pump bought as an insurance spare; used only if the working pump fails; useful life 6 years                                                                                | 2,40,000  |
| 5   | Consumable spares (lubricants, bearings) used during the year                                                                                                                                              | 60,000    |
| 6   | On 1 April 2025, a new conveyor added, increasing capacity by 30%; to be used over the remaining life of the structure                                                                                     | 12,00,000 |

**Required:** (a) [8 marks] Classify each event as capital or revenue expenditure with reasons and state the accounting treatment. (b) [7 marks] Compute the depreciation for FY 2025-26 and the total charge to profit or loss.

### SOLUTION

#### (a) Classification

| No. | Treatment                                                  | Reason                                                                                                                                                                                |
|-----|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Capitalise ₹30,00,000; <b>derecognise</b> old lining       | It is the replacement of a significant component. The old lining (cost ₹24,00,000, life 5 years) is removed at its carrying amount of ₹4,80,000, which is charged as a <b>loss</b> .  |
| 2   | Revenue – expense                                          | Day-to-day servicing keeps the plant in working condition; it does not add benefits.                                                                                                  |
| 3   | Capitalise as a component; <b>derecognise</b> old overhaul | A major overhaul that restores the plant for the next 3 years meets the recognition criteria. The previous overhaul (₹7,50,000) is fully depreciated at 30 Sept 2025, so nil is lost. |
| 4   | Capitalise as PPE                                          | A major spare or standby item expected to be used over <b>more than one period</b> qualifies as PPE. Depreciate over 6 years.                                                         |
| 5   | Revenue – expense (inventory when consumed)                | Consumables are used within a year; they are stores and spares consumed in production.                                                                                                |
| 6   | Capitalise                                                 | Capacity is increased by 30%, so <b>future economic benefits increase</b> . Depreciate over the remaining life of the structure (8 years).                                            |

#### Loss on derecognition of the old furnace lining

| Particulars                                    | ₹               |
|------------------------------------------------|-----------------|
| Cost of old lining                             | 24,00,000       |
| Accumulated depreciation, 4 years × ₹4,80,000  | (19,20,000)     |
| <b>Carrying amount = loss on derecognition</b> | <b>4,80,000</b> |

**(b) Depreciation for FY 2025-26**

| Component                         | Working                                                     | Depreciation (₹) |
|-----------------------------------|-------------------------------------------------------------|------------------|
| Main structure                    | $96,00,000 \div 12$ (carrying $64,00,000 \div 8$ remaining) | 8,00,000         |
| New furnace lining                | $30,00,000 \div 5$ years (from 1 April 2025)                | 6,00,000         |
| Old overhaul (up to 30 Sept 2025) | $7,50,000 \div 3 \times 6/12$                               | 1,25,000         |
| New overhaul (from 1 Oct 2025)    | $9,00,000 \div 3 \times 6/12$                               | 1,50,000         |
| Standby pump                      | $2,40,000 \div 6$ years                                     | 40,000           |
| Conveyor                          | $12,00,000 \div 8$ remaining years                          | 1,50,000         |
| <b>Total depreciation</b>         |                                                             | <b>18,65,000</b> |

**Total charge to profit or loss**

| Particulars                                 | ₹                |
|---------------------------------------------|------------------|
| Depreciation                                | 18,65,000        |
| Loss on derecognition of old furnace lining | 4,80,000         |
| Routine repairs and maintenance             | 4,00,000         |
| Consumable spares                           | 60,000           |
| <b>Total</b>                                | <b>28,05,000</b> |

**Final answer**

Depreciation ₹18,65,000; total charge to profit or loss ₹28,05,000 (including loss of ₹4,80,000 on the old lining).

**Why? – Simple explanation**

Ask one question for every cost incurred after the asset was bought: **does it create or extend future benefits beyond what was originally expected, or does it just maintain the existing level?** If it creates or extends benefits, capitalise; if it maintains them, expense.

Whenever you capitalise a **replacement**, the old part you are replacing must come off the books, because you cannot own the same part twice.

A standby spare is capital only if it is a **major item** expected to last more than one year; everyday consumables are expenses.

**Watch out – Common mistakes / Exam tip**

- Do not depreciate the old lining in FY 2025-26; it was removed on 1 April 2025.
- Do not forget the 6 months of depreciation on the old overhaul before it is fully written off.
- The new overhaul is a **separate component** with its own 3-year life; it is not spread over the plant's life.



**Q11. Exchange of Assets and Deferred Payment**

**Level 3 • Multi-concept (exam pattern) | 15 marks** | Concepts tested: exchange with and without commercial substance, gain on exchange, cash price equivalent, interest on deferred payment

**PROBLEM****(a) Exchange with commercial substance [5 marks]**

Alpha Ltd. exchanges an old machine (cost ₹20,00,000; accumulated depreciation ₹12,00,000; fair value ₹9,00,000) for a new machine of Beta Ltd. and pays **₹2,00,000 in cash**. The fair value of the new machine is ₹11,00,000. The exchange **has commercial substance**.

**(b) Exchange without commercial substance [5 marks]**

Same facts as (a), but the exchange **lacks commercial substance**.

**(c) Deferred payment [5 marks]**

On 1 April 2025 Alpha bought another machine. The cash price is **₹25,00,000**. Alpha will pay **₹30,25,000 at the end of 2 years** (nothing before). Useful life 10 years, nil residual value, straight-line. Find the cost, the interest for each year and the journal entries.

**SOLUTION****(a) With commercial substance**

| Particulars                                                                                 | ₹                |
|---------------------------------------------------------------------------------------------|------------------|
| Carrying amount of old machine (20,00,000 – 12,00,000)                                      | 8,00,000         |
| Fair value of old machine given up                                                          | 9,00,000         |
| Gain on exchange (9,00,000 – 8,00,000)                                                      | 1,00,000         |
| <b>Cost of new machine = fair value of asset given up + cash paid (9,00,000 + 2,00,000)</b> | <b>11,00,000</b> |

| Particulars                             | Dr (₹)    | Cr (₹)    |
|-----------------------------------------|-----------|-----------|
| New machine A/c                         | 11,00,000 |           |
| Accumulated depreciation A/c            | 12,00,000 |           |
| To Old machine A/c                      |           | 20,00,000 |
| To Bank A/c                             |           | 2,00,000  |
| To Gain on exchange (Profit & Loss) A/c |           | 1,00,000  |

**(b) Without commercial substance**

| Particulars                                                           | ₹                |
|-----------------------------------------------------------------------|------------------|
| Carrying amount of old machine                                        | 8,00,000         |
| Add: cash paid                                                        | 2,00,000         |
| <b>Cost of new machine = carrying amount of asset given up + cash</b> | <b>10,00,000</b> |
| Gain or loss                                                          | Nil              |

| Particulars                  | Dr (₹)    | Cr (₹)    |
|------------------------------|-----------|-----------|
| New machine A/c              | 10,00,000 |           |
| Accumulated depreciation A/c | 12,00,000 |           |
| To Old machine A/c           |           | 20,00,000 |
| To Bank A/c                  |           | 2,00,000  |

**(c) Deferred payment**

Cost of the machine = **cash price equivalent = ₹25,00,000**. The extra ₹5,25,000 (30,25,000 – 25,00,000) is interest, implied rate = 10% p.a. ( $25,00,000 \times 1.1 \times 1.1 = 30,25,000$ ).

| Year                  | Opening liability (₹) | Interest @ 10% (₹) | Closing liability (₹) |
|-----------------------|-----------------------|--------------------|-----------------------|
| FY 2025-26            | 25,00,000             | 2,50,000           | 27,50,000             |
| FY 2026-27            | 27,50,000             | 2,75,000           | 30,25,000             |
| <b>Total interest</b> |                       | <b>5,25,000</b>    |                       |

| Particulars                                         | Dr (₹)    | Cr (₹)    |
|-----------------------------------------------------|-----------|-----------|
| <b>1 April 2025</b>                                 |           |           |
| Machine A/c                                         | 25,00,000 |           |
| To Deferred payment liability A/c                   |           | 25,00,000 |
| <b>Each year-end: depreciation (25,00,000 ÷ 10)</b> |           |           |
| Depreciation A/c                                    | 2,50,000  |           |
| To Accumulated depreciation A/c                     |           | 2,50,000  |
| <b>31 March 2026</b>                                |           |           |
| Finance cost A/c                                    | 2,50,000  |           |
| To Deferred payment liability A/c                   |           | 2,50,000  |
| <b>31 March 2027</b>                                |           |           |
| Finance cost A/c                                    | 2,75,000  |           |
| To Deferred payment liability A/c                   |           | 2,75,000  |
| Deferred payment liability A/c                      | 30,25,000 |           |
| To Bank A/c                                         |           | 30,25,000 |

**Final answer**

(a) Cost ₹11,00,000, gain ₹1,00,000. (b) Cost ₹10,00,000, no gain. (c) Cost ₹25,00,000; interest ₹2,50,000 and ₹2,75,000; depreciation ₹2,50,000 a year.

**Why? – Simple explanation**

When an asset is swapped for another with **commercial substance** (the business really changes in risk, cash flows or value), the new asset is recorded at fair value, and the difference between fair value and the old asset's carrying amount is a real gain or loss.

When the swap has **no commercial substance**, nothing has really changed economically, so no gain is recognised. The new asset simply takes over the carrying amount of the old one (plus any cash paid).

When payment is delayed beyond normal credit terms, the asset is recorded at its **cash price**. The extra you pay is interest for the delay and is recognised over the credit period.

**Watch out – Common mistakes / Exam tip**

- In (a) the measure is the fair value of the asset **given up** unless the fair value of the asset received is more clearly evident.
- If neither fair value can be measured reliably, the exchange is also measured at the carrying amount of the asset given up.
- Interest on a machine that is bought ready to use is expensed; interest on a **qualifying asset** under construction may be capitalised (see Q9).

**Q12. Revaluation, Transfer of Surplus, Disposal and Disclosure**

**Level 3 · Multi-concept (exam pattern) | 15 marks** | Concepts tested: revaluation, depreciation on revalued amount, transfer to retained earnings, derecognition, disclosures

**PROBLEM**

Meridian Ltd. bought a machine on **1 April 2022** for **₹30,00,000** (useful life 10 years, nil residual value, straight-line). From **31 March 2025** it follows the revaluation model. On that date an **independent valuer** fixed the fair value at **₹24,50,000**; remaining life 7 years. The company transfers the excess depreciation from revaluation surplus to retained earnings each year. On **30 September 2026** the machine was sold for **₹20,00,000** net of selling costs.

**Required:** (a) [5 marks] Revaluation entry at 31 March 2025 (elimination method), depreciation and transfer for FY 2025-26. (b) [5 marks] Depreciation, gain or loss and treatment of the surplus on sale in FY 2026-27. (c) [5 marks] Disclosures required for FY 2025-26.

**SOLUTION****(a) Revaluation and FY 2025-26**

| Particulars                                                                          | ₹                |
|--------------------------------------------------------------------------------------|------------------|
| Carrying amount at 31 March 2025 (30,00,000 – 3 × 3,00,000)                          | 21,00,000        |
| Fair value                                                                           | 24,50,000        |
| <b>Revaluation surplus (to OCI)</b>                                                  | <b>3,50,000</b>  |
| Depreciation FY 2025-26 = 24,50,000 ÷ 7                                              | 3,50,000         |
| Depreciation on original cost = 30,00,000 ÷ 10                                       | 3,00,000         |
| <b>Excess depreciation transferred from revaluation surplus to retained earnings</b> | <b>50,000</b>    |
| <b>Carrying amount at 31 March 2026 (24,50,000 – 3,50,000)</b>                       | <b>21,00,000</b> |
| <b>Revaluation surplus at 31 March 2026 (3,50,000 – 50,000)</b>                      | <b>3,00,000</b>  |

| Particulars                      | Dr (₹)   | Cr (₹)   |
|----------------------------------|----------|----------|
| <b>31 March 2025</b>             |          |          |
| Accumulated depreciation A/c     | 9,00,000 |          |
| To Machine A/c                   |          | 9,00,000 |
| Machine A/c                      | 3,50,000 |          |
| To Revaluation surplus (OCI) A/c |          | 3,50,000 |
| <b>31 March 2026</b>             |          |          |
| Depreciation A/c                 | 3,50,000 |          |
| To Accumulated depreciation A/c  |          | 3,50,000 |
| Revaluation surplus A/c          | 50,000   |          |
| To Retained earnings A/c         |          | 50,000   |

**(b) Sale on 30 September 2026**

| Particulars                                                            | ₹                |
|------------------------------------------------------------------------|------------------|
| Carrying amount at 1 April 2026                                        | 21,00,000        |
| Less: depreciation for 6 months $(3,50,000 \times 6/12)$               | (1,75,000)       |
| <b>Carrying amount at date of sale</b>                                 | <b>19,25,000</b> |
| Net sale proceeds                                                      | 20,00,000        |
| <b>Gain on sale (to profit or loss)</b>                                | <b>75,000</b>    |
| Revaluation surplus at 1 April 2026                                    | 3,00,000         |
| Less: excess depreciation for 6 months $(50,000 \times 6/12)$          | (25,000)         |
| <b>Balance of surplus transferred to retained earnings on disposal</b> | <b>2,75,000</b>  |

| Particulars                                          | Dr (₹)    | Cr (₹)    |
|------------------------------------------------------|-----------|-----------|
| Depreciation A/c                                     | 1,75,000  |           |
| To Accumulated depreciation A/c                      |           | 1,75,000  |
| Bank A/c                                             | 20,00,000 |           |
| Accumulated depreciation A/c $(3,50,000 + 1,75,000)$ | 5,25,000  |           |
| To Machine A/c                                       |           | 24,50,000 |
| To Gain on disposal of PPE (Profit & Loss) A/c       |           | 75,000    |
| Revaluation surplus A/c $(25,000 + 2,75,000)$        | 3,00,000  |           |
| To Retained earnings A/c                             |           | 3,00,000  |

**(c) Disclosures for FY 2025-26 (extract)**

| Reconciliation of carrying amount of machine                       | ₹                |
|--------------------------------------------------------------------|------------------|
| Carrying amount at 1 April 2025 (revalued)                         | 24,50,000        |
| Additions / disposals / impairment                                 | Nil              |
| Depreciation for the year                                          | (3,50,000)       |
| <b>Carrying amount at 31 March 2026</b>                            | <b>21,00,000</b> |
| Gross carrying amount 24,50,000; accumulated depreciation 3,50,000 |                  |

| Revaluation disclosures                                                         | Detail                                      |
|---------------------------------------------------------------------------------|---------------------------------------------|
| Effective date of revaluation                                                   | 31 March 2025                               |
| Independent valuer involved?                                                    | Yes                                         |
| Carrying amount had the cost model been used                                    | 18,00,000 $(30,00,000 - 4 \times 3,00,000)$ |
| Revaluation surplus: opening ₹3,50,000; transfer to retained earnings (₹50,000) | Closing ₹3,00,000                           |

Also disclose: measurement basis used (revaluation model for this class), depreciation method (straight-line), useful life (remaining 7 years at revaluation date), methods and significant assumptions used in estimating fair value, and any restrictions on title or pledges.

Check: carrying amount on cost basis ₹18,00,000 + surplus ₹3,00,000 = revalued carrying amount ₹21,00,000. ✓

**Final answer**

(a) Surplus ₹3,50,000; depreciation ₹3,50,000; transfer ₹50,000 a year. (b) Gain **₹75,000** in profit or loss; ₹3,00,000 surplus goes to retained earnings, **not** through profit or loss. (c) Carrying amount **₹21,00,000**; cost-model carrying amount ₹18,00,000; surplus ₹3,00,000.

**Why? – Simple explanation**

Depreciation on a revalued asset is higher than on the old cost. The extra depreciation eats into profit, but the surplus is really being **realised** through use. So each year an equal amount is moved from revaluation surplus to retained earnings, without touching profit or loss.

When the asset is sold, the remaining surplus is also **realised** and moves to retained earnings. The gain or loss on sale is calculated only from the **carrying amount**, so it does not include the surplus.

Disclosures let readers see what the numbers would have been under cost, who valued the asset, and how the surplus moved.

**Watch out – Common mistakes / Exam tip**

- The transfer of surplus to retained earnings is a movement within equity. It is **never** routed through profit or loss.
- Charge depreciation up to the sale date before computing the gain.
- In practice deferred tax would also be recognised on a revaluation surplus (Ind AS 12); it is ignored here.